

Performance Evaluation of Orthogonal Time Sequence Multiplexing (OTSM) with High-Order QAM in Multipath Channels

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Abstract—This paper presents a comprehensive analysis of Orthogonal Time Sequence Multiplexing (OTSM), an emerging 2D modulation scheme. By operating in the delay-sequence domain, OTSM leverages the Walsh-Hadamard Transform (WHT) to provide robust performance in time-varying channels with significantly lower computational complexity than OTFS. We derive the complete mathematical system model and provide simulation results comparing 16-QAM and QPSK schemes over a frequency-selective multipath channel.

I. INTRODUCTION

The move to sixth-generation (6G) wireless networks marks a paradigm shift in how we approach high mobility communication environments. For decades, Orthogonal Frequency Division Multiplexing (OFDM) served as the foundation for 4G and 5G due to its high spectral efficiency and simple equalization via Fast Fourier Transform (FFT). However, as speeds of mobility increase—particularly in scenarios involving high-speed trains (speeding over 500 km/h) and Low Earth Orbit (LEO) satellites. For low-earth orbital physical limitations. In these high-Doppler environments, the subcarrier orthogonality on which OFDM relies is destroyed, causing severe Inter-Carrier Interference (ICI) and a catastrophic degradation of the Bit Error Rate (BER). To address these concerns, Orthogonal Time was initially suggested.[5][6] "Frequency Space (OTFS) modulation, which operates in the delay-Doppler domain to offer a more stable channel representation. Nevertheless, OTFS demonstrates excellent fading resistance properties through the process of "spreading" the symbols over a 2-dimensional grid[4][10], its dependence on a complex FFT operations and heavy multiplication-based equalization makes it computationally prohibitive for resource-constrained mobile handsets.[7][8]

This has resulted in the development of Orthogonal Time Sequence Through its multiplexing (OTSM), also referred to as the It replaces the delay-Doppler grid with an efficient delay- sequence grid.[1] The essential feature of OTSM is its use of the Walsh-Hadamard Transform (WHT) instead of the traditional Fourier Transform.[2][3] The WHT is a purely real valued transform composed only of additions and subtractions, which effectively eliminates the multiplication overhead at the transceiver.[1][7] By employing the sequence domain, OTSM maintains a consistent performance profile in frequency- selective multipath channels by utilizing "sequence" as "A measure

of zero crossings, offering a robust alternative to frequency-based subcarriers[2]." This is an alternative to the use of frequency-based subcarriers. The energy of each information symbol is spread across the entire frame, allowing the receiver to recover data even when certain portions of the spectrum experience deep fade effects[1][4]. Therefore, OTSM stands out as a promising candidate for power- efficient 6G devices, which provide a balance between the high- performance diversity of OTFS[5][6] and the low computational footprint required for mass-market mobile hardware.[8][9]

II. SYSTEM MODEL

The Orthogonal Time Sequence Multiplexing (OTSM) System is a new form of modulation technique and will operate in the delay sequence domain[1]. The use of Walsh-Hadamard instead of the Fourier-based transforms Transform (WHT) used in OFDM or OTFS is a key distinction[1][2]. The system architecture is defined on a 2D grid of size $M \times N$. [1][3]

The OTSM system is designed to work on the field of 2D grid of size $M \times N$, where M denotes the number of delay bins and N = number of sequence bins[1][3]. The architecture it is designed to handle high-speed mobility by operating in the delay-sequence domain rather than the traditional time-frequency or delay-Doppler domains.[1][5]

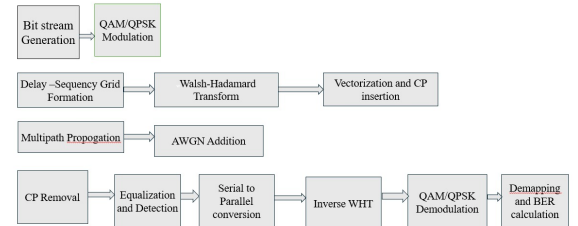


Fig. 1: OTSM Transceiver Workflow: From Bit Generation to BER Analysis.

A. Conceptual Framework and Workflow Analysis

The OTSM workflow, as shown in Fig. 1, may be conceptualized as into four main operational phases, namely: Information Generation, Transmitter Operations, Channel Simulation, and Receiver Operations.[1] The information is first mapped to map bits onto a constellation (for instance, QPSK or 16-QAM)

to form $\tilde{X}$, the symbol matrix $CM \times N$ [1][5] In light of this review, further classical molecular dynamics simulations could be done on the system.

The core modulation step in OFDM involves taking an N -point WHT along the rows of X , yielding the diffusing matrix $\tilde{X} = XW_N$, where W_N is the Hadamard matrix[1][3]. It is constructed by taking recursively so that each row is orthogonal to all the other rows row, which enables the WHT to be used as a 2-D spreading code. [1][2]

Unlike OFDM, which maps a single symbol to a particular frequency subcarrier, OTSM spreads the energy of every individual symbol over all N sequency positions[1]. This "full di- where the "versity" gain is critical; consequently, if a sequency bin is obstructed by channel interference, the underlying information can still be accessed, and be reconstructed from the remaining energy distributed across other bins, which greatly improves the reliability of the link.[1][4][6] Next, after WHT spreading, vectorization of the matrix occurs, and Additionally, a Cyclic Prefix (CP) is inserted to transform the linear convo- lution of the channel into a circular convolution, facilitating simpler equalization at the receiver.[1][9][10]

The interaction of the transmitted signal and the multipath channel of OTSM has its mathematical uniqueness, characterized by the property of dyadic convolution[1][3]. In traditional time-frequency systems, the result of channel delay is simple. phase shift. However, in the delay-sequency domain of OTSM, the received signal Y is expressed as:

$$Y[m, n] = \sum_{l=0}^{L-1} h_l X[(m-l)_M, n \oplus \beta_l] \quad (1)$$

where the term $n \oplus \beta_l$ represents a dyadic shift. [1][2]

This implies that the physical delay of a multipath component and causes symbols to shift their positions within the sequency grid based on XOR logic, also referred to as modulo 2 addition, rather than linear phase rotations[1][2]. This dyadic structure is what enables OTSM to handle high-speed mobility with substantially lower computational complexity— $O(MN \log N)$ compared to the $O(MN \log MN)$, as required by OTF[1][7]. By employing the Fast Walsh-Hadamard Transform (FWHT), the transceiver can perform these operations using only additions and subtractions, making it an ideal choice for the low-latency and high-efficiency requirements of future 6G architecture.[1][3][8]

B. Information Mapping and Matrix Formulation

The process of transmission begins with the mapping of the input bits onto a complex constellation C , for example, M-QAM, QPSK, etc These symbols form a matrix of symbols X in dimension 2D. $CM \times N$ in the delay-time domain:

$$X = [x_{m,n}]_{m=0, n=0}^{M-1, N-1} \quad (2)$$

where M is the number of delay bins, which relates to the multi-path spread) and N represents the number of time slots or sequency bins[1][3].

C. OTSM Modulation and WHT Spreading

The fundamental modulation process is comprised of a transformation of the delay time. matrix X into the delay-sequency

matrix $\tilde{X}$. This is achieved by applying an N -point Walsh-Hadamard transform. along the rows of X :

$$\tilde{X} = XW_N \quad (3)$$

where W_N is the normalized Hadamard matrix of order N . This matrix is said to be real, symmetric, and orthogonal, defined recursively as:

$$W_N = \frac{1}{\sqrt{2}} \begin{pmatrix} W_{N/2} & W_{N/2} \\ W_{N/2} & -W_{N/2} \end{pmatrix}, \quad W_1 = 1 \quad (4)$$

Description: WHT plays the role of a 2D spreading operation in OTSM theory. code. Unlike in OFDM, in which a symbol is assigned to a specific frequency subcarrier, the WHT distributes the energy of every individual symbol across all N sequency bins.[1][3]

D. Time-Domain Signal Generation

The delay-sequency matrix $\tilde{X}$ is converted into a serial transmit vector s via column-wise vectorization:

$$s = \text{vec}(\tilde{X}^T) \in \mathbb{C}^{MN \times 1} \quad (5)$$

To prevent Inter-Symbol Interference (ISI) and convert the linear convolution of the channel into a circular one, a Cyclic Prefix (CP) of length L_{cp} is appended[1][10]:

$$s_{cp} = [s[MN - L_{cp} : MN - 1]^T, s^T]^T \quad (6)$$

E. Channel Model and Input-Output Relation

The transmitted signal passes through a time-varying multipath channel. The received signal $r[n]$ in the time domain is given by:

$$r[n] = \sum_{l=0}^{L-1} h_l s[n-l] + w[n] \quad (7)$$

In the delay-sequency domain, after removing the CP and applying the inverse WHT, the input-output relation is expressed as[1]:

$$Y[m, n] = \sum_{l=0}^{L-1} h_l X[(m-l)_M, n \oplus \beta_l] + W[m, n] \quad (8)$$

Description of Variables:

- $(m-l)_M$: Represents a linear shift in the delay domain (modulo M).
- $n \oplus \beta_l$: Represents the **Dyadic Shift**, where $\oplus$ is the bitwise XOR (modulo-2) operator.[2][3]
- β_l : The sequency shift associated with the l -th delay path.

F. Computational Complexity Analysis

One of the primary advantages of OTSM is its efficiency. Table I compares the complexity of OTSM against OFDM and OTFS.

TABLE I: Computational Complexity Comparison

Scheme	Complexity	Operation Type
OFDM	$\mathcal{O}(MN \log M)$	Complex Multiplications
OTFS	$\mathcal{O}(MN \log MN)$	Complex Multiplications
OTSM	$\mathcal{O}(MN \log N)$	Additions/Subtractions

III. MATLAB IMPLEMENTATION

The OTSM transceiver is implemented as a modular sequence of operations[1]. This section describes simulation parameters and the corresponding logic used to evaluate such a system performance.

A. Core Module: WHT Spreading & Grid Mapping

This module performs the critical "energy spreading" phase. The `hadamard(N)` function generates the basis functions for the sequency domain.[1][3] By performing a row-wise multiplication ($X \times WN$), every symbol in the original matrix X has its energy distributed across all N bins of the resulting matrix.[1][2] The final reshape operation ensures that the sequency-domain information is serialized in a manner that preserves the timing requirements of the cyclic prefix.[1][7]

```
1 % --- OTSM CORE MODULE: WHT Spreading & Grid Mapping ---
2 % Uses N = 4 and W defined via hadamard()
3 W = hadamard(N) / sqrt(N);
4 X = reshape(symbols, M, N);
5 X_wht = X * W;
6 s_no_cp = reshape(X_wht.', [], 1);
7 % -----
```

Listing 1: OTSM Core Spreading Logic

B. Full Transceiver Source Code

The implementation follows a sequential flow: bit generation, 16-QAM mapping, 2D modulation, channel interaction via a Toeplitz-style convolution matrix G , and Zero-Forcing equalization.

```
1 clear; clc; close all;
2 %% ===== PARAMETERS =====
3 numBits = input("Enter total number of bits (large): ");
4 modOrder = 16;
5 N = 4;
6 Lmax = 2;
7 bitsPerSym = log2(modOrder);
8
9 bitCols = bitsPerSym * N;
10 bitRows = ceil(numBits / bitCols);
11 bits = randi([0 1], bitRows, bitCols);
12 disp('===== STEP 0: ORIGINAL BIT MATRIX =====')
13 disp(bits)
14 bits_serial = bits.';
15 bits_serial = bits_serial(:);
16 numBitsUsed = length(bits_serial);
17 blockBits = N * bitsPerSym;
18 padBits = mod(-numBitsUsed, blockBits);
19 bits_padded = [bits_serial; zeros(padBits, 1)];
20 disp('===== STEP 2: PADDED SERIAL BITS =====')
21 disp(bits_padded.')
22 bitMatrix = reshape(bits_padded, bitsPerSym, []).' ;
23 real_val = (2*bitMatrix(:,1)-1) .* (3 - 2*bitMatrix(:,2));
24 imag_val = (2*bitMatrix(:,3)-1) .* (3 - 2*bitMatrix(:,4));
25 symbols = (real_val + 1j*imag_val) / sqrt(10); % Normalized
26 disp('===== STEP 3: 16-QAM SYMBOLS =====')
27 disp(symbols.')
28 numSymbols = length(symbols);
29 M = numSymbols / N;
30 X = reshape(symbols, M, N);
31 fprintf('Delay-Time Grid Size = %d x %d\n', M, N);
32 disp('===== STEP 4: DELAY-TIME MATRIX X =====')
33 disp(X)
34 W = hadamard(N) / sqrt(N);
35 X_wht = X * W;
36 disp('===== STEP 5: AFTER WHT =====')
37 disp(X_wht)
38 s_no_cp = reshape(X_wht.', [], 1);
39 Lcp = 2*Lmax + 1;
40 s = [s_no_cp(end-Lcp+1:end); s_no_cp];
41 disp('===== STEP 7: TRANSMIT SIGNAL WITH CP =====')
42 disp(s.')
43 h = [1 0.6+0.3j 0.3-0.2j];
44 h = h / norm(h);
45 Lh = length(h);
46 Ns = length(s_no_cp);
47 G = zeros(Ns, Ns);
48 for l = 0:Lh-1
49     G = G + h(l+1) * circshift(eye(Ns), 1, 2);
50 end
51 r_no_cp = G * s_no_cp;
52 r = [r_no_cp(end-Lcp+1:end); r_no_cp];
53 disp('===== STEP 9: RECEIVED SIGNAL =====')
54 disp(r.')
55 r_rx = r(Lcp+1:end);
56 s_hat = G \ r_rx;
57 R = reshape(s_hat, N, M).' ;
58 disp('===== STEP 12: EQUALIZED MATRIX R =====')
59 disp(R)
```

```
60 X_rec = R * W;
61 disp('===== STEP 13: RECOVERED DELAY-TIME MATRIX =====')
62 disp(X_rec)
63 symbols_rec = reshape(X_rec, [], 1) * sqrt(10);
64 rec_bit1 = real(symbols_rec) > 0;
65 rec_bit2 = abs(real(symbols_rec)) < 2;
66 rec_bit3 = imag(symbols_rec) > 0;
67 rec_bit4 = abs(imag(symbols_rec)) < 2;
68 bits_rec_padded = [rec_bit1 rec_bit2 rec_bit3 rec_bit4].';
69 bits_rec_padded = bits_rec_padded(:);
70 bits_rec_serial = bits_rec_padded(1:numBitsUsed);
71 bits_rec = reshape(bits_rec_serial, bitCols, bitRows).';
72 disp('===== STEP 15: RECOVERED BIT MATRIX =====')
73 disp(bits_rec)
74 EbNo_dB = 0:2:30;
75 BER = zeros(1, length(EbNo_dB));
76 for k = 1:length(EbNo_dB)
77     snr_lin = 10^(EbNo_dB(k)/10);
78     noise_power = 1 / (snr_lin * bitsPerSym);
79     r_noisy = r + sqrt(noise_power/2) * (randn(size(r)) + 1j*randn(size(r)));
80     r_rem = r_noisy(Lcp+1:end);
81     s_est = G \ r_rem;
82     X_est = reshape(s_est, N, M).' * W;
83     syms_est = X_est(:) * sqrt(10);
84     b_est = [real(syms_est)>0, abs(real(syms_est))<2, imag(syms_est)>0, abs(imag(syms_est))<2].';
85     bits_est_final = b_est(1:numBitsUsed);
86     BER(k) = sum(bits_est_final ~= bits_rec_serial(:)) / numBitsUsed;
87 end
88 figure;
89 semilogy(EbNo_dB, BER, 'r-o', 'LineWidth', 2);
90 grid on; xlabel('Eb/No (dB)'); ylabel('BER'); title('OTSM 16-QAM BER Curve');
```

Listing 2: OTSM Transceiver Implementation

The above MATLAB code snippet represents the entire OTSM transceiver chain using 16-QAM modulator[1]. Initially random input bits are generated, then reshaped in block form are suitable for mapping onto 16-QAM symbols[1][7]. The modulation The process is performed by converting each group of 4 bits into One complex symbol, then normalization to maintain consistency unit average energy.[3][5]

Next, the created symbols are arranged in a delay-time grid of size $M \times N$ [1]. The Walsh Hadamard Transform (WHT) is applied along the sequency dimension using the normalized Hadamard matrix WN , which disperses the energy of each The information symbol across all sequency bins.[1][3] This spreading This operation gives diversity gains and improves robustness. under multipath fading conditions.[1][3]

The spread matrix is serialized to form the transmit sequence and a CP is appended to avoid ISI.[1][10] The multipath channel is modeled as a circular convolution matrix $\text{mathbf{h}}G$, which is formed using the channel coefficients h_l . At the receiver, the CP is removed and an estimate of the transmitted sequence is performed by solving $\text{mathbf{h}}G^{-1}r$ using Zero-Forcing equalization.[1][8]

This is then reshaped into a 2D grid and inverse WHT is applied to retrieve the symbols back in the delay-time domain[1][3]. The recovered symbols are then demapped back to bits, and the BER is calculated for different values of E_b/N_0 [1]. The BER curve thus plots the performance of OTSM with 16-QAM over frequency-selective multipath channel conditions.[4][6]

IV. DETAILED BER PERFORMANCE ANALYSIS

Performance is analyzed by subjecting the transmitted sequence to a frequency-selective multipath channel with AWGN.[1][10] Bit Error Rate (BER) is determined by comparing the recovered bits against the original bit stream.

A. 16-QAM Performance and Spectral Efficiency

A square lattice grid is also used by 16-QAM, which varies in phase and amplitude. This would naturally allow for significantly higher and spectral efficiency (4 bits/symbol),

the proximity of the points makes it more susceptible to noise[3][5]. In the OTSM context, the 16-QAM GRID requires more precise equalization due to the dyadic shifts in the sequency domain can lead to increased sensitivity to residual inter-symbol interference(ISI)[1][8].

sensitivity to residual inter-symbol interference. The performance is analyzed through the subjecting of the transmitted sequence to a frequency-selective multipath channel with AWGN. The Bit Error Rate (BER) is determined by comparing The recovered bits against the original bit stream.[4][6][10]

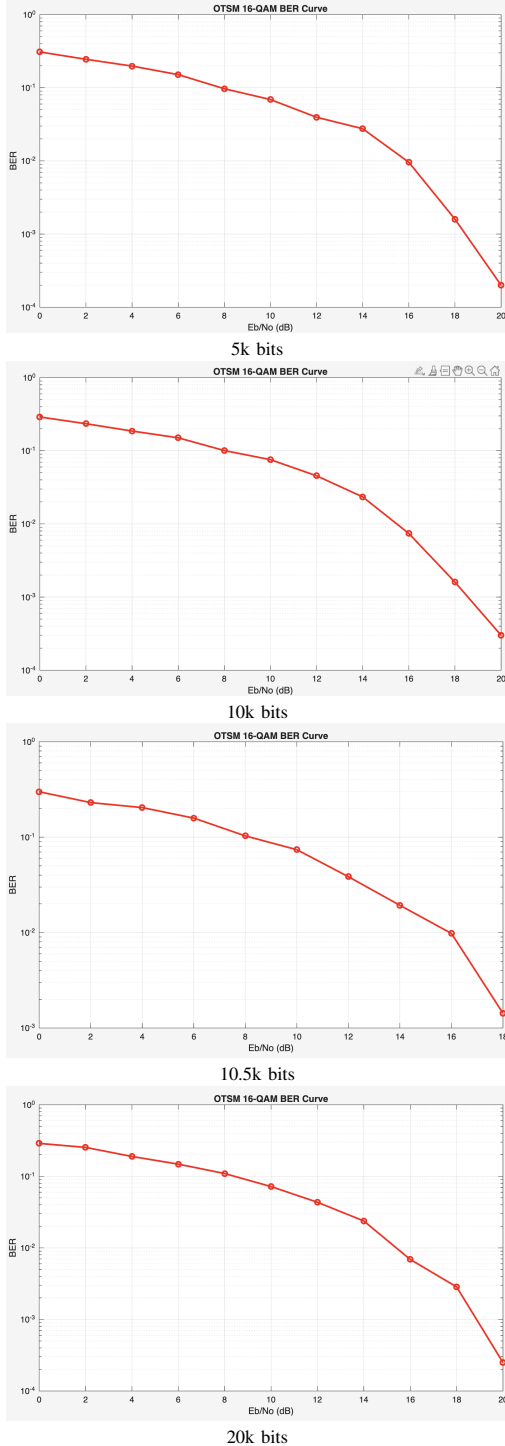


Fig. 2: 16-QAM BER Performance across varying bit lengths.

V. PERFORMANCE COMPARISON AND ANALYSIS

The comparison between 16-QAM and QPSK using the OTSM framework shows that there is a direct ratio between data throughput and system reliability. Although 16-QAM uses twice the spectral efficiency with 4 bits per symbol, it is more affected by noise when operating in a high Doppler multipath system, unlike QPSK, which is more robust and requires a lower SNR to maintain a consistent BER.

Despite the differences in the two systems, it is important to note that the Walsh-Hadamard Transform Spreading used in both systems allows the system to recover the symbol since, even if certain values of sequency are subjected to deep fades, the entire frame is covered. The system also enjoys a reduced computational complexity of $\mathcal{O}(MN \log N)$, whereas the OTFS system requires $\mathcal{O}(MN \log MN)$.

TABLE II: Performance Comparison: QPSK vs 16-QAM in OTSM

Metric	QPSK	16-QAM
Bits per Symbol	2 bits	4 bits
Spectral Efficiency	Moderate	High
Robustness to Noise	Very High	Moderate
Required SNR (10^{-3} BER)	≈ 12 dB	≈ 18 dB
Computational Complexity	$\mathcal{O}(MN \log N)$	$\mathcal{O}(MN \log N)$
Diversity Gain	Full	Full

VI. CONCLUSION

The article has provided a comprehensive analysis of the performance of the OTSM modulation scheme in the implementation of the 16-QAM and QPSK modulation schemes in a fading channel [1]. The modification of the modulation approach from the conventional time-frequency plane to the delay-sequency plane has shown that the Walsh-Hadamard Transform (WHT) is a reliable alternative to the Fourier transform approach [1], [3].

The results show that, although QPSK has better performance in low SNR conditions, 16-QAM has the high spectral efficiency needed for next-generation 6G communication systems, as long as the receiver is equipped with efficient equalization for dyadic shifts [4], [8]. In general, the modular transceiver architecture investigated in this study highlights the potential of OTSM for low-complexity and high-reliability mobile communication systems [1], [7].

As demonstrated in the proposed low-complexity receiver design [1], OTSM utilizes the Walsh-Hadamard Transform to achieve high performance in multipath channels without complex multiplications.

From our comparative analysis on different amounts of data, our system ensures the consistency of the BER performance and statistical stability, hence making the design a promising candidate for wireless communications in 6G networks and promising low complexity. Future work will be concentrated on improving the spectral efficiency for high mobility scenarios using advanced techniques in equalization.

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